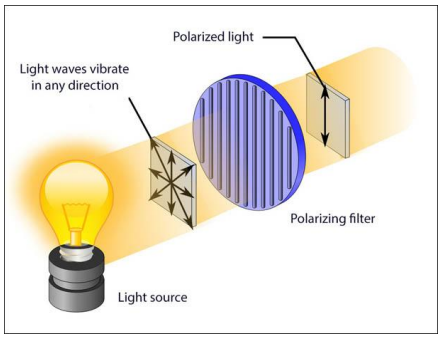
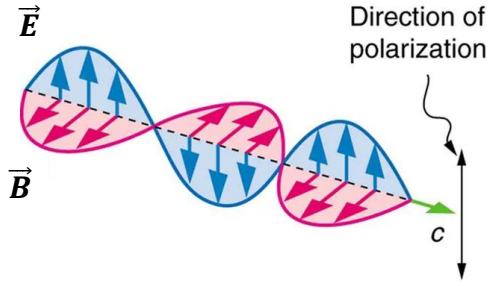
Chapter 56 – EM Wave Polarization
Calculus Physics II

Wave Polarization – The process by which the vibrations of a transverse wave are selected to lie in a preferred plane. EM waves can be polarized; the electric fields of the EM wave all lie in the same plane and the magnetic field lines in the same plane but perpendicular to the electric field.





- For an electromagnetic wave the polarization is defined by the electric field $\vec{E}$ and is when the electric field vibration always occurs along one direction (same plane) the EM wave is said to be polarized.
- The magnetic field is always perpendicular to the electric field and would also be polarized.

Unpolarized light – A wave in which the vibrations of the $\vec{E}$-field occur randomly in all directions. Light emitted by most common light sources are un-polarized, the electric field direction of each ray of light is randomly oriented to the others, there is no one direction or plane of oscillation.

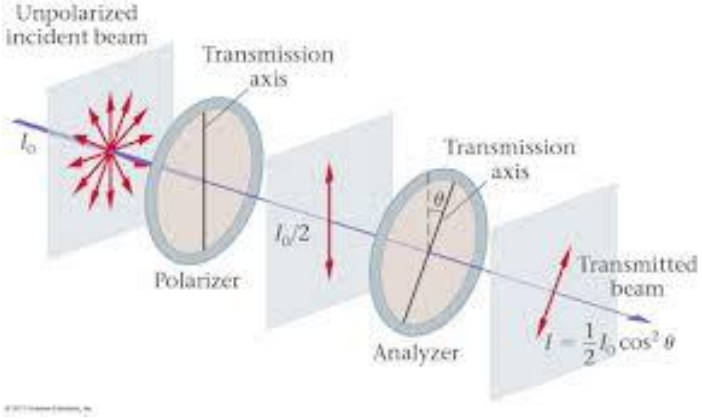
Linear Polarized light – A wave in which the electric field vibrations only occur along one plane or direction.

Polarizer – is a material that selectively absorbs/reflects/refracts/scatters the components of the electric field that are parallel to the Transmission axis of the polarizer leaving a portion of the transmitted/reflected light polarized.

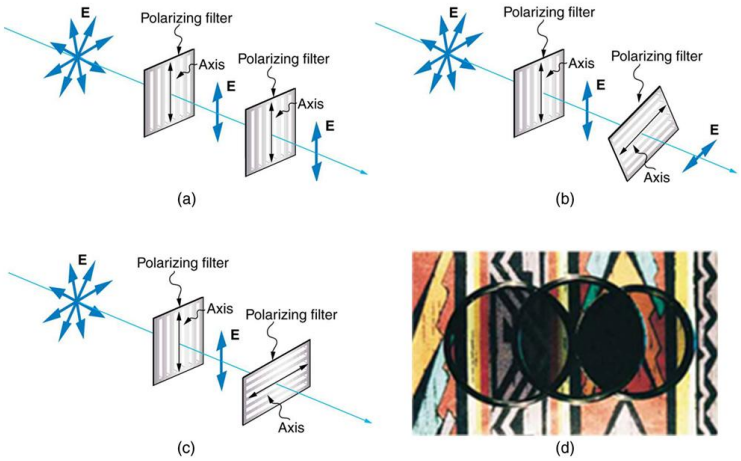
Transmission axis of a polarizing material is the preferred transmission of an electric field passing through the material. Only the component of the electric field vector that is along the transmission axis makes it through the polarizer, the other component is either absorbed or reflected by the polarizer.

- Light that is unpolarized and incident on a single polarizer the intensity of the light transmitted by the polarizer will not change with rotation of the polarizer, but the polarization direction changes with the rotation of the transmission axis.

Analyzer – A single polarizer used to determine the state of polarization of incident light. The intensity of the transmitted light will vary as the analyzer's transmission axis rotates from parallel to the Electric field (maximum transmission) to perpendicular to the field (minimum transmission) for polarized incident light.



Intensity of Light before and after Polarizer – The transmitted intensity of the light through a polarizer will depend on if the light starts out polarized and the orientation of the polarized direction of the light relative to the polarizer's transmission axis.



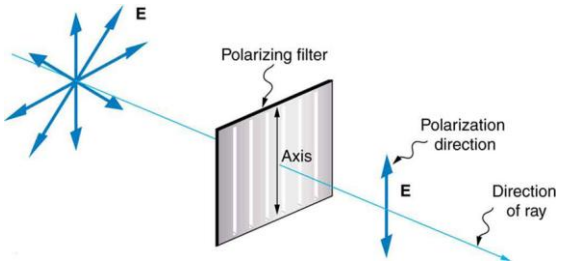
(a) Unpolarized light is incident on the first polarizer with its transmission axis vertical, the light that exits the first polarizer will be polarized vertical and have half the intensity as that of the incident light. The vertically polarized light then is incident on a second polarizer (analyzer) that has its transmission axis vertical. The incident polarized light will not be affected by the second polarizer because there is no angle between the transmission axis of the polarizer and the polarization of the incident light.

(b) Unpolarized light is incident on the first polarizer with its transmission axis vertical, the light that exits the first polarizer will be polarized vertical. The vertically polarized light then is incident on a second polarizer (analyzer) that has its transmission axis at some angle to the polarized light that is incident on it. The intensity of the incident polarized light will be reduced by the second polarizer because there is an angle between the transmission axis of the polarizer and the polarization of the incident light.

(c) Same as before, except the second polarizer's transmission axis is perpendicular to the incident lights polarization, no light will be transmitted through the second polarizer. This is known as "crossed polarizers".

Unpolarized light incident on polarizer – For unpolarized incident light the total sum of all the components of the electric field vector for all the EM waves are equally divided along the transmission axis and perpendicular to the transmission axis, so only half the intensity of light gets though when the incident light is unpolarized.

$$I = \left(\frac{1}{2}\right) I_o$$



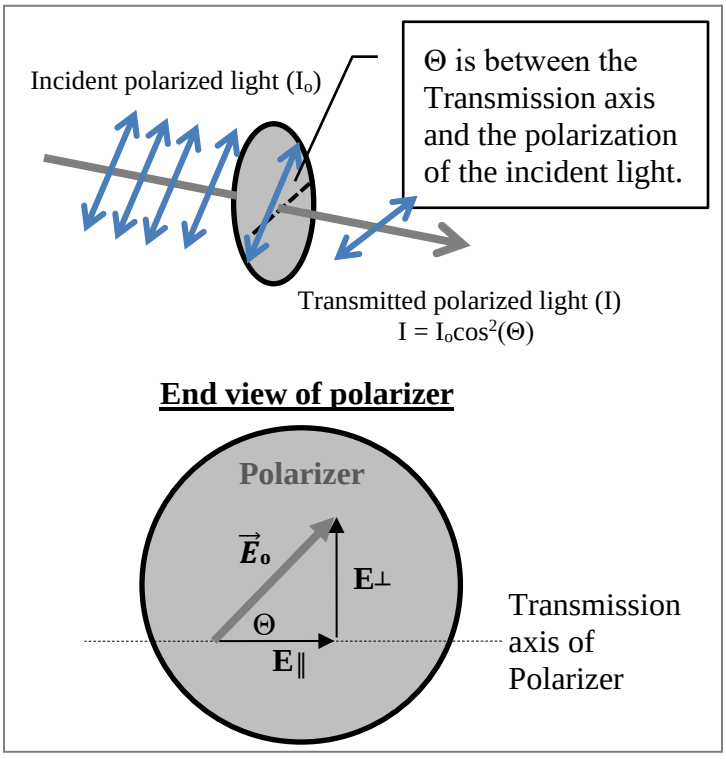
I_o - Intensity of light before polarizer.
I - Intensity of light after polarizer.

Example: Unpolarized light with an intensity of 100 W/m^2 is incident on a polarizer with its axis of polarization along the vertical axis, what is the intensity of the light come out of the polarizer and what direction is the light polarized?

$I = (½)I_o \quad = 50$ W/m^2

- The light will be polarized in the vertical direction along the transmission axis.

Polarized light incident on a polarizer – The transmitted intensity depends on the orientation of the polarization direction of the light relative to the transmission axis of the polarizer. The intensity of the transmitted light will vary as the analyzer's transmission axis rotates from parallel to the Electric field (maximum transmission, $\Theta = 0°$) to perpendicular to the field (no transmission, $\Theta = 90°$) for polarized incident light.



$\vec{E}_o$ is the incident electric field vector, Θ is the angle between the axis of <u>transmission axis of the polarizer</u> and the <u>polarization plane of the incident EM wave</u>.

E$_{\perp}$ is the component of the Electric field vector that is perpendicular to the transmission axis of the polarizer and is not transmitted.

E$_{\parallel}$ is the component of the Electric field vector that is parallel to the transmission axis of the polarizer and is totally transmitted through the polarizer.

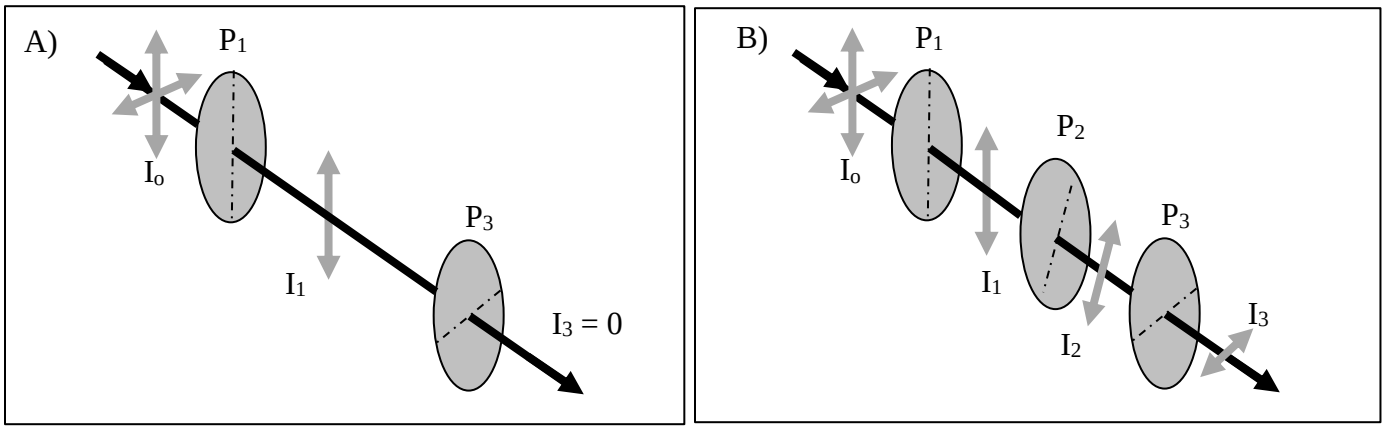
<u>Malus's Law</u> – When <u>polarized light</u> is incident on polarizer where I_o is the incident intensity, and I is the transmitted intensity, the transmitted intensity depends on the angle between the electric field (polarization direction) of the light and the transmission axis of the polarizer.

$$I = I_o\cos^2\Theta = I_o(\cos\Theta)^2$$

Where

Θ is the angle between the Electric Field of the polarized light and the Transmission axis of the polarizer.
I_o is the polarized incident EM wave intensity on the polarizer before it reaches the polarizer.
I is the transmitted EM wave intensity after going through the polarizer.

<u>Light transmitted through crossed polarizer's</u> – the first diagram shows two crossed polarizers' with unpolarized light incident on Polarizer 1 (P_1), the light is polarized perpendicular to the transmission axis of polarizer 3 (P_3) so no light is transmitted through P_3. If we place # 2 polarizer between P_1 and P_3 that has its transmission axis at some angle to P_1's axis but not perpendicular, then light will get through P_3 because now there is a component of the electric field along the transmission axis of P_3.



Example: Looking **at diagram A)** if the incident light I_o = 1,200 W/m^2 what is the intensity of light that exits P_1 and P_3 if P_1's transmission axis is 90° to P_3's axis?

I_o is unpolarized light incident on P_1 therefore
$I_1 = ½I_o$ = 600 W/m^2 polarized vertically

I_1 is vertically polarized, and it is incident on a polarizer with its horizontal transmission axis therefore the angle between the polarization of the light incident and the transmission axis of the polarizer is 90°

$I_3 = I_1(\cos\Theta)^2 = (600\ W/m^2)(\cos(90°) = 0\ W/m^2$

Diagram B) if the incident light I_o = 1,200 W/m^2, if P_1 has its axis of transmission vertical, P_2 has its axis of transmission at 30° to the vertical and P_3 has its axis of transmission at 90° from the vertical then what is the intensity of light that is exiting P_1, P_2 and P_3?

I_o is unpolarized light incident on P_1 therefore
$I_1 = ½I_o$ = 600 W/m^2 polarized vertically

I_1 is vertically polarized, and it is incident on a polarizer that with its axis of transmission at 30° to the vertical therefore

$I_2 = I_1(\cos\Theta)^2 = (600\ W/m^2)(\cos(30°) = 450\ W/m^2$
and polarized 30° from the vertical

I_2 is polarized 30° from the vertically and it is incident on a polarizer that with its axis of transmission at 90° from the vertical therefore the angle between the polarized light and the axis of transmission of P_3 is $\Theta = 90° - 30° = 60°$

$I_3 = I_2(\cos\Theta)^2 = (450\ W/m^2)(\cos(60°) = 112.5\ W/m^2$

- There is light getting through the third polarizer due to the presence of the polarizer in the middle, if the middle polarizer was not there then there would be crossed polarizers, and no light is transmitted.

Applications using polarized light.

Why use polarized sunglasses? – Light that is reflected off a surface is either partially or completely polarized (Brewster's Angle), reflected light is sometimes called glare and keeps you from seeing clearly, with polarized sunglasses, the axis of transmission of the polarizer's are oriented to block this reflected (partially polarized) light thereby blocking the glare.

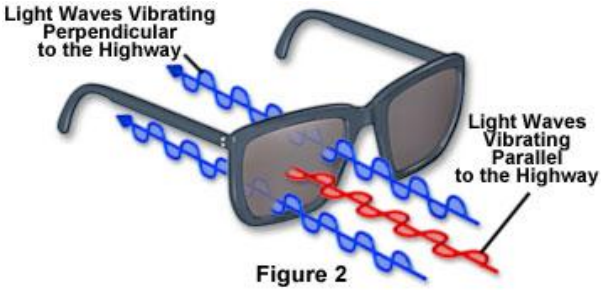
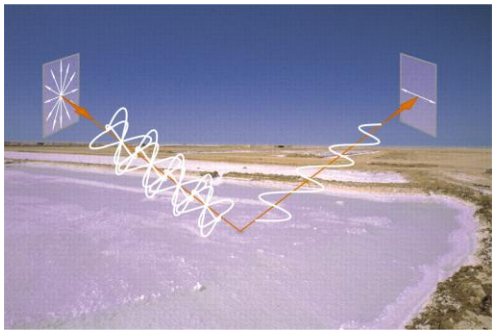


Figure 2



Stretched Polymers – Liquid Crystal Displays (LCD) – The liquid crystal is sandwiched between two crossed polarizers' so that polarized light enters the liquid crystal. By applying a voltage to the liquid crystal the polarization of the light traveling though it can be rotated. With no voltage applied the light goes through the LC and is blocked by the other polarizer and there is nothing displayed but, when the voltage is applied the polarizer rotates the polarization of the light and it goes through the second polarizer and the segment is lit up.

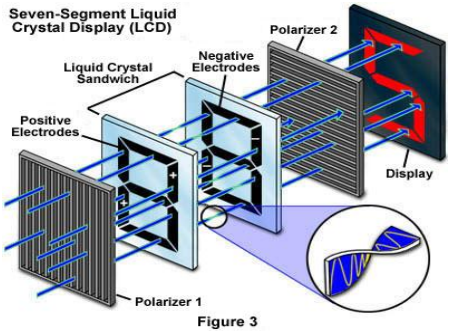
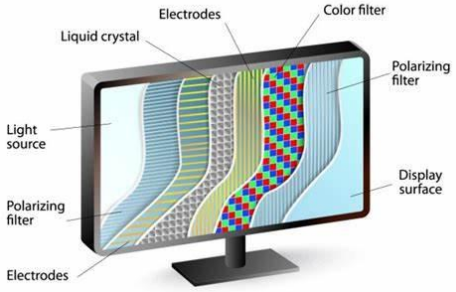
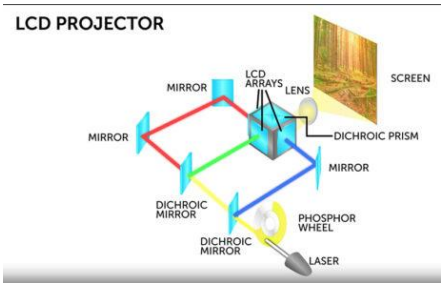


Figure 3

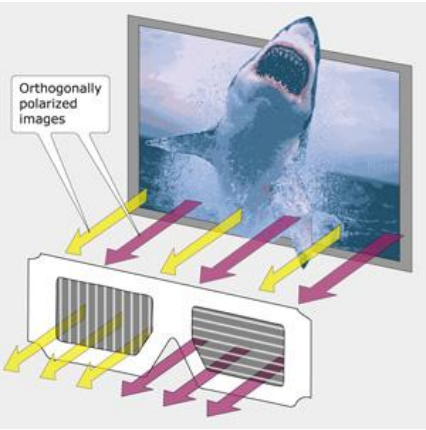


- If you are wearing polarized sunglasses sometimes the displays will appear dark, if you tilt your head with the glasses on the image will change.

Digital Light Processing (DLP) projectors do not inherently require polarization for image formation, as they use tiny mirrors (a DMD chip) to reflect or redirect light. However, polarization can be used with DLP projectors for 3D applications and for combining light beams.



3D movies and polarized glasses – Two images of the same scene are projected or displayed with the two images having perpendicular polarizations and slightly different viewpoints. When wearing polarized glasses what have perpendicular polarizers, each eye sees a slightly different image and only that image. The mind interprets the two slightly different perspectives as a three-dimensional image and that's what is seen.



Strain detectors using polarization – crossed polarizer's – stress produces a strain in material and in transparent material it will rotate the polarization of the light that goes through it and this rotation depends on the frequency (color) of the light. The more strain in the material, the larger the polarization rotation and the greater the change in colors. Below is a piece of plastic that was placed between crossed polarizers, the plastic rotates the polarization of the light by different amounts depending on the strain in the different regions of the material.



